Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
5	*(a)*	(i)	Use of BA = flux or Faraday's law $\Delta(BA)/t$ (1) [or by impl.] $\frac{2.1\times(610-120)\times10^{-4}}{0.016}$ seen (or similar) (1)	1	1		2	2	2
		(ii)	Anticlockwise (accept words if not in diagram) (1) FRHR or Right hand screw rule or FLHR on electrons (1) **Alternative (for explanation)** Lenz's law: [direction of induced flux must oppose so out of paper.]	1	1		2		2
	(b)		Resistance = $\frac{\rho l}{\pi r^2}$ quoted or used ($0.0034\,\Omega$) (1) Use of $I=\frac{V}{R}$ (1) $\frac{6.4\times\pi\times0.0015^2}{2.65\times10^{-8}\times0.91}$ [=1880 A] or equivalent seen (1)	1	1 1		3	3	3
	(c)		$V=\pi r^2 l$ used [= 6.43×10^{-6} m^3] (1) Mass = density × volume used [=0.0174 kg] (1) Energy = IVt used (or equivalent 195 J) (1) Energy = $mc\Delta T$ used (1) Answer = 12.5 K so Iestyn wrong(1)			5	5	5	5
	(d)		Any 2 × (1) from • Consent of patient discussed [e.g. volunteers] (1) • possible injury to patient discussed [e.g. long-term effects] (1) • Appropriateness of experiment carried out on model / dead body / animal instead discussed (1) Conclusion Argument is logical and leads to a conclusion (even if a mixed conclusion) (1)			3	3		
			Question 5 total	**3**	**4**	**8**	**15**	**10**	**12**